ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 A ✓ 1.3 C ✓ 1.4 D ✓ 1.5 A ✓ 1.6 B ✓ 1.7 C ✓ 1.8 B ✓ 1.9 A ✓ 1.10 D ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 C ✓ 3.2 F ✓ 3.3 I ✓ 3.4 H ✓ 3.5 K ✓ 3.6 B ✓ 3.7 D ✓ 3.8 A ✓ 3.9 G ✓ 3.10 J ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 The total number of magnetic lines of force in a magnetic field (unit: weber) ✓ (1)

4.1.2 The amount of magnetic flux passing through one square metre at right angles to the flux (unit: tesla) ✓ (1)

4.1.3 How easily a material allows magnetic flux to pass through it ✓ (1)

4.1.4 A coil wound on a soft-iron core that becomes a magnet only while current flows through it ✓ (1)

4.2.1 $\text{mmf} = \text{N} \times \text{I} = 400 \times 2{,}5 \checkmark = 1\,000\ \text{At} \checkmark$ (2)

4.2.2 $\text{H} = \text{mmf} \div \text{l} = 1\,000 \div 0{,}2 \checkmark = 5\,000\ \text{At/m} \checkmark$ (2)

4.3 First law: an emf is induced in a conductor whenever it cuts magnetic lines of force, or the flux linking a coil changes ✓✓. Second law: the size of the induced emf is proportional to the rate at which the flux is cut or changes ✓✓. (4)

4.4 $\text{e} = \text{B} \times \text{l} \times \text{v} \checkmark = 0{,}8 \times 0{,}3 \times 5 \checkmark = 1{,}2\ \text{V} \checkmark$ (3)

[15]

QUESTION 5

5.1.1 $1 \div R_P = 1 \div 10 + 1 \div 15 + 1 \div 30 \checkmark = 6 \div 30 \checkmark; R_P = 5\ \Omega \checkmark$ (3)

5.1.2 $R_T = 5 + 5 = 10\ \Omega \checkmark$ (1)

5.1.3 $I = V \div R = 24 \div 10 \checkmark = 2{,}4\ A \checkmark$ (2)

5.1.4 $V = I \times R = 2{,}4 \times 5 \checkmark = 12\ V \checkmark$ (2)

5.2.1 $N_s = N_p \times V_s \div V_p \checkmark = 1\,150 \times 12 \div 230 \checkmark = 60\ \text{turns} \checkmark$ (3)

5.2.2 $I_p = I_s \times V_s \div V_p = 2 \times 12 \div 230 \checkmark = 0{,}10\ A \checkmark$ (2)

5.3 Any FOUR ✓✓✓✓: yoke (frame); field poles and field windings; armature (core and windings); commutator; brushes; shaft; bearings; end shields. (4)

5.4 In a series motor the field winding is connected in series with the armature ✓; in a shunt motor the field winding is connected in parallel (across) the armature ✓. (2)

5.5 Any ONE ✓: electric trains / trams; cranes and hoists; starter motors in vehicles; electric drills – where a high starting torque is needed. (1)

[20]

QUESTION 6

6.1 $R = \rho L \div A \checkmark = 0{,}028 \times 10^{-6} \times 500 \div (2{,}5 \times 10^{-6}) \checkmark\checkmark = 5{,}6\ \Omega \checkmark$ (4)

6.2 $R_t = R_0(1 + \alpha_0 t) \checkmark = 20(1 + 0{,}00427 \times 60) \checkmark = 25{,}12\ \Omega \checkmark$ (3)

6.3 Correct symbol with the arrow on the emitter pointing outwards ✓; base ✓; collector and emitter labelled ✓ (3)

6.4 The Arduino pin is connected through a base resistor to the base of the transistor ✓; when the pin goes HIGH, base current flows and the transistor saturates (switches fully on), so current flows from the 12 V supply through the relay coil to ground ✓; when the pin goes LOW, the transistor cuts off and the relay switches off (a diode across the coil protects the transistor) ✓. (3)

6.5 $I_E = I_C + I_B$ ✓✓ (2)

[15]

QUESTION 7

7.1 Any SIX in a logical order ✓✓✓✓✓✓: switch off the circuit; set the meter to the correct current range (start on the highest range / A or mA); put the red lead in the correct current socket and the black lead in COM; break the circuit where the current is to be measured; connect the meter in series so all the current flows through it; switch on and read the value; switch off, remove the meter and reconnect the circuit; return the leads and selector to the voltage position. (6)

7.2 Correct NAND symbol (AND shape with a small circle at the output) ✓; truth table ✓✓✓✓: (5)

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

7.3 Two-input AND gate with inputs A and B ✓; NOT gate on input C ✓; two-input OR gate ✓ with the AND output ✓ and the NOT output as its inputs; output labelled X ✓ (5)

7.4 Any FOUR ✓✓✓✓: keep it clean and dry (wipe with a dry or slightly damp cloth, not a wet one); store it in its case; remove the batteries when it will not be used for a long time; switch it off after use; check the leads for damage before use; do not drop it or expose it to heat; have it calibrated regularly; use the correct range. (4)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE